Medical Report Generation using Chest X-Ray

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***Abstract*— Advances in the domain of Deep Learning, Natural Language Processing, Natural Language Generation, significant advances in computing power and the availability of large datasets has led to encouraging results in the field of medical imaging and the analysis of medical reports. Chest X-rays is an important modality that is used by radiologists around the world for managing the diagnosis and treatment of a large number of ailments in patients. Automatic generation of the radiology reports can help the radiologists in coping up with their workload and can also minimize the rate of incorrect diagnosis and overlooked diagnosis. The studies lead us to the realization that Natural Language Processing and Natural Language Generation have a lot of potential as applied to medical imaging. However, lots of work needs to be further carried out for mitigating the challenges involved so that the automatic reports become trustworthy, and therefore acceptable by the medical fraternity.**

Keywords*— *Medical report generation, Chest X-Rays, Natural Language Processing, Convolutional Neural Networks

I. Introduction

Chest X - rays is the most commonly prescribed examination by the radiologists for screening, diagnosing and managing a large number of conditions in the patients. Natural Language Processing (NLP) combines artificial intelligence, linguistics and computer science to allow machines to read and understand human language for meaningful and useful purposes. [12] Text can be converted into structured representation with the help of NLP and it also helps in enabling automatic identification and extraction of information. NLP can help in processing large amounts of data and its applications may improve with standardizing task-based performance requirements, report standardizations and external validation.

The superimposition of nearby anatomical structures along the direction of projection makes the interpretation of the chest x-rays a difficult task. As a result, there are variations in the analysis of the x-ray images when observed by different radiologists. Most often, the guidelines are based on irregular datasets without taking the historical data into account. Discrepancies in the clinical definitions and procedures is also a concern. There can be issues of not considering all the variables involved, thereby impacting the diagnosis. To create a robust NLP system, first establishing the knowledge base with an acceptable amount of structured data is a must.

There are several obstacles to overcome when it comes to importing handwritten notes and recording vocal dictation, etc. [11]. Accurately and quickly identifying the information contained in the radiology reports has a good potential in becoming a decision - support system for the radiologists and also in identifying the patients who need urgent care.

Deep learning in combination with Natural Language Processing can help enhance medical reasoning, prioritize the evidence and the results, and it can become a ubiquitous tool for providing better health care for all.

II. Literature Review

The authors [1] have employed the transformer model to generate radiology reports and believe that it outperforms competitive benchmarks when measured by the metrics used for language generation as well as the report coherence. The authors have utilized the MIMIC-CXR dataset that contains 227, 835 imaging reports along with 377,110 total images for 65, 379 patients. The dataset was constrained to radiology reports that contained findings columns and the rest of the data was divided into training (70%), validation (10%) and testing datasets (20%). The data was divided on the basis of patient ID instead of on specific radiology reports to avoid issues due to multiple radiology tests conducted by the subjects. For neural machine translation, the authors employed a transformer model, which comprises an encoder-decoder model. The encoder was made to operate directly on image features instead of operating on word embeddings. A pretrained DenseNet-121 model was used to extract the features from the chest radiographs. The decoder was used in a similar manner to its original neural machine translation setting. The continuous - bag-of - words Word2Vec method was used to pre-train word embeddings for all the words occurring at least five times in the training corpus. The model performance was analyzed for the metrics CIDER, METEOR, ROUGE-L, BLEU. The model was also analyzed and fine-tuned for the recall, precision and the F1 score for positive assessments. [1]

The authors [2] have used a LSTM model to generate radiology reports. They present a new model called Vispi, which annotates an image by visually identifying and localizing the typical thoracic diseases and then creates a report using the attentive LSTM model. The Indiana University (IU) X-ray dataset, which is an publicly available dataset with 3955 radiology reports associated with chest X-rays, has been used for the purpose of this study. The dataset was constrained to radiology reports that consisted of the findings columns and the rest of the data was divided into training (70%), validation (10%) and testing datasets (20%). The authors have gone about this report generation by initially describing an integrated image interpretation framework that gives disease annotation and medical report generation. The authors employed massive datasets to enhance medical image translation while training on a small number of reports. They assessed the applicability of NLP performance metrics for medical report generation and, lastly, proved the usefulness of using a heatmap to localize critical results in an X-ray. The training data here is the IU X-ray dataset. The disease classification and localization module are built on a 121-layer DenseNet which is pre-trained with Chest X-ray8 dataset. Grad-CAMs were

used for the heatmap. The report generation uses attentive LSTM model which works on an encoder-decoder model. A pre-trained ResNet 101 model was used for the encoder. The model performance was analyzed for the metrics CIDEr, ROUGE, BLEU-1, BLEU-2, BLEU-3, BLEU-4. And a ROC Curve was plotted to compare results of TieNet and Vispi. [2]

The authors [3] trained a Deep Convolutional Neural Network (DCNN) model using a Tuberculosis (TB)-specific chest x-ray (CXR) dataset and assessed it on a non-TB-specific CXR dataset from another source. The NIH ChestX-ray8 database, HIPAA-compliant datasets, and the National Library of Medicine (NLM) Shenzhen No.3 Hospital X-ray collection were used in this research. They divided the dataset into three categories: training (75%), intramural test (10%) and validation (15%) sets. Using Inception V3, a pre-trained DCNN, they used color-space augmentation, crop-flip, and rotational algorithms to classify each image as having TB characteristics or as healthy. The DCNN model that resulted was used to estimate the prevalence of TB-associated CXR in the ChestX-ray8 dataset. Researchers used receiver operating characteristic (ROC) curves and areas under the curve (AUC) to evaluate model performance and establish the appropriate cut point for TB detections . The predictive performance of the Inception V3 TensorFlow model in the Shenzhen dataset was maintained in this study for training, validation, and intramural test images. Higher resolution pictures may enhance accuracy, especially for subtle results, and more study is needed in this area.

The authors [4] have implemented a layered convolutional neural network model to generate radiology reports. Their model named CheXnet returned a report accompanied with a heatmap limiting the areas indicative of thoracic diseases though mostly concerned with detecting pneumonia. The authors utilized the ChestX-ray14 dataset, which includes 112,120 frontal view X-ray images from 30,805 people.The dataset was constrained to radiology reports that contained findings columns and the rest of the data was divided into training (70%), validation (10%) and testing datasets (20%). The data was divided on specific radiology reports avoiding the issues of a patient overlap. The authors used Class Activation Maps (CAMs) for generation of heatmaps which help in localizing and classifying the pathologies. The CheXnet is a 121-layer DenseNet pretrained on ImageNet. The model performance was analyzed for the AUROC metric. The model was also analyzed and compared with four practicing radiologists, the average F1 score of the model far exceeded the radiologists' average F1 score.

In [5], the authors have used a To produce medical reports for different views of chest X-ray pictures, a Multilevel Multi-Attention (MLMA) encoder-decoder system based on integrating Context Level Visual Attention and Textual Attention was used. To achieve the objective the author has used the IU Chest X-ray dataset. The dataset comprises 3955 radiological reports of different forms with 7470 multiple views of Chest X-rays. TRepeated sentences are replaced with a single sentence; images with no report and inaccurate words are removed. 500 samples were used for testing and 500 samples were chosen at random for validation in order to assess performance.

The VGG16's block5 pool layer encoded the images. All reports were processed in advance to eliminate unnecessary words, numbers, and punctuation. The Nadam (Nesterov-Adam) optimizer was used to train all proposed models to minimize cross-entropy with an initial learning rate of 0.001. To train the end-to-end MLMA model, the initial batch of input had a size of 128. The adaptive beam search approach was utilized for testing to create the best initial word sequentially. The author compared the Bleu-1, Bleu-2, Bleu-3, Bleu-4, Meteor, Rouge-L, and CIDER models to the data. The author's MLMA model yields excellent results for unknown X-ray images; according to the author, this strategy beats state-of-the-art methods for the development of medical reports for any chest X-ray.

The authors [6] utilized the MIMIC-CXR dataset. MIMIC-CXR contains 473,057 chest x-ray pictures and 206,563 reports. The authors have tested various language generation models to evaluate and understand the NLG metrics. The models that were tested for the evaluation were Conditional n-gram language model, CNN-RNN model, Nearest neighbor (1-NN) and random retrieval baseline amongst many. The CNN-RNN model used a LSTM based encoder-decoder approach for extraction of the output with captions. This approach was used to minimize the cross-entropy loss. The decoding mechanism was beam search. The authors have taken the help of CheXpert during the evaluation process to assess the models for accuracy and precision. The models were analyzed for the BLEU and CIDEr metrics. The authors concluded that the CNN-RNN model and 1-NN model top the linguistic and clinical accuracy table. In the BLEU and CIDEr metrics, the CNN model with beam search decoding outdid the other models. The authors are of the opinion that adding additional decoding techniques yields benefits for neural models and better image extractors. The authors have concluded that the standard NLG metrics give a skewed output which favors irrelevant reports based on the number of grammatical blunders.

The authors [7] utilized the Indiana University (IU) X-ray dataset, an accessible image dataset containing 3955 radiology reports matched with chest X-rays, for their investigation. The dataset was restricted to radiology reports with findings columns, and the rest of the data was split into training (70%), validation (10%), and testing datasets (20 percent). For their study, the authors used an LSTM model. They have developed a new approach for assessing the trained chest X-ray report production, in which the encoder is employed for visual feature extraction and the learned features are used to assess their performance for thoracic disease categorization. CheXNet is used to classify thoracic disease, which in turn employs DenseNet-121. The Chest X-ray 14 dataset was utilized for this. Chest X-ray 14 is one of the largest publicly accessible radiograph datasets, including 112,120 frontal view chest X-rays of 30,805 distinct individuals. In this case, each picture is tagged with up to 14 different thoracic pathology tags taken from the appropriate report. The tags are accurate more than 90% of the time. The samples are randomly selected as 70% for training, 10% for validation and 20% for testing purposes. The report generator first uses CNN-RNN model to process

the input image followed by Bi-LSTM model. The pre-trained model ResNet-152 has been used for building the image encoder model. The report decoder uses a stacked 2-layer LSTM model. The model performance was analyzed for the metrics BLEU, METEOR, ROUGE and CIDEr.

The authors [8] concentrated on producing clinically accurate reports. Their system model produces a list of topics. The system creates the final report based on each topic. To improve readability, the system is fine-tuned using reinforcement learning. The authors have tested this model on the MIMIC-CXR dataset and Open-I dataset. They have introduced the hierarchical strategy of generation with a CNN-RNN-RNN architecture. Evaluation metrics used by authors for Automatic LG are CIDEr-D, ROUGE-L, and BLEU. The authors have also compared their model with some baseline models like 1-NN, Show and Tell (S&T), Show, Attend and Tell (SA&T), and TieNet. After testing different models, the authors observed that their NLG model that promptly helps to optimize the CIDEr score, and can reach a much remarkable performance in terms of the metrics of NLP.

The authors [9] have worked on the Indiana University chest x-ray dataset (IU-Xray) consisting of 7430 images from 3825 individuals. The authors modified a transformer-based model that had previously been trained and used GPT2 to train it. This model works on an encoder-decoder approach where The encoder is a CheXNet model that has already been trained. used for extracting the image information and the decoder is used for generating the full-text reports. For the encoder part of the model the authors tweaked the CheXnet model which is a DenseNet121 trained on the Chest X ray 14 dataset by the addition of a new final layer. The decoder consisted of a pre-trained distilGPT2 because of its adaptability and speed compared to the normal GPT2 model. This model was compared with 4 different models on the basis of NLG, BLEU, METEOR, ROGUE-L and Cider metrics. The model's results were also cross-checked by an expert radiologist and the results for the normal ones were almost 99% accurate but the abnormal ones needed some improvement. The time taken by the model to generate the report turned out to be faster when compared to existing models. In addition, the authors argue that traditional word-overlap-based techniques are inadequate for systems like this. This model puts heavy emphasis on the semantic feature similarities for better reports.

The authors [10] suggested a framework for adversarial reinforced report generation for chest x-ray images. The Adversarial Reinforced framework focuses on diagnostic accuracy and language fluency. The authors used the Indiana University Chest X-ray dataset, which consists of 7470 frontal and lateral view images combined with 3955 diagnostic reports. Each report is divided into four sections: comparison, indication, findings, and impression. Authors were focused on generating the findings. They have used the PyTorch Python library to implement all the proposed models. To verify the performance of the proposed approach, the authors have used four competitive models viz. CNN-RNN, CoAtt, MvH-Attl-MC and NLGR-CCR. For evaluation of language fluency, the authors have used Natural Language Generation viz. BLEU-4, METEOR, ROUGE-L, and CIDEr. The CheXpert labeler was used to produce reports for diagnostic accuracy, and the precision, recall, and F1 scores were determined.

III. Background

In medical imaging, X-rays which are a form of electromagnetic radiation are used. In order to recommend the correct diagnosis to the patients, the radiologists must analyze the X-ray properly. The most common radiologic examination ordered is a chest X-ray. This image helps your doctor determine whether you have heart disease, a collapsed lung, pneumonia, fractured ribs, emphysema, cancer, or any of a variety of other conditions. Preparing medical-imaging reports can be difficult for unskilled radiologists and pathologists, especially those operating in remote locations with insufficient healthcare quality; on the other side, writing imaging reports can be monotonous and time consuming for experienced radiologists and pathologists.

IV. Proposed System

A. Design Details

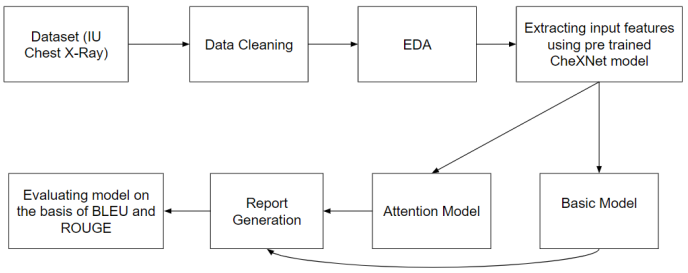

Fig. 1. Flow Diagram of the Project

B. Methodology

For this project we have used various mechanisms and methodologies. Below are the descriptions of the same. During Preprocessing of Data we have used various python libraries like Numpy,Pandas,Natural Language Toolkit Library for text data type and for images we have libraries like PIL and Computer Vision. For Data Analysis we have used EDA i.e Exploratory Data Analysis process.We have used this process because it helps us to visualize common traits or anomalies in our data.For visualization we have used python libraries like matplotlib.pylot and matplotlib.image and Seaborn. Further for extracting image features/vectors from images we have used Transfer Learning Mechanism and for that we have used the CheXNet pre-trained model. After extraction of image vectors we created train,test and validation data set using sklearn library from Python and we also did tokenization using the Tokenizer module from Natural Language Toolkit Library. Now we move on to creating models. First we created a Basic Encoder-Decoder model. We used the Tensorflow.Keras library from Python. Then we created an Attention model to improve our result for that we used Bahdanau Additive Attention Mechanism.

C. Implementation

1. Preprocessing of Data

Preprocessing of data is a very important step for any data related projects. In this step we clean the data i.e we remove redundancy,null values and gibberish values from our datasets. For this project we have used IU Chest X-ray dataset [15] which contains 7471 images with both front view and lateral view of each patient's chest. Around 3955 patient reports are available. Every image in the dataset has four captions namely Indication, Comparison, Findings and Impressions. These give a better description of the salient entities and events. The finding caption gives maximum information present in images.

2. Exploratory Data Analysis

We obtain insights about the data after properly analyzing the data and can make decisions based on these insights.So the process of Exploratory Data Analysis is done to get some key insights about the data.This process enables us to get information like the maximum and minimum number of images associated with a report,most frequently used words in the report. Most of the reports are associated with 2 images. This process also helps us to find any anomalies in our datasets.

3. Feature Extraction

Before creating any model we need to be careful about what kind of data we are feeding into the model. Feature Extraction is a technique we have used to extract input features from our image dataset and then divide it into training,test and validation dataset.

4. Encoder - Decoder Model

Encoder - Decoder model is a sequence-to-sequence deep learning model that takes a set of items (in this case, image characteristics) and produces a new set of items (reports). The encoder examines each item in the input sequence and aggregates the data into a vector known as the context. The encoder delivers the context to the decoder after processing the complete input sequence, and the decoder begins producing the output sequence item by item. For precise tuning, we used the Dense and Dropout layers in the encoder component of our model. For better output, we used an additional embedding layer and an LSTM layer in the decoder.

5. Attention Model

The Attention model [14] is an improvisation of the Encoder-Decoder model. We have used Bahdanau Additive Attention Mechanism here. The Attention can be stated as a vector of important weights. The encoder here is on similar lines as the Encoder-Decoder model while the decoder has some additions. The context vector in this case is produced by the Attention layer after getting the encoder output and decoder input. Then the initial state of the previous decoder and the context vector is passed through the RNN for generating the output. Dropout layer is used for tuning and regularization of the model. The Beam Width Search algorithm has been used here in order to predict the most probable impression of the images.

D. Comparison of Models

TABLE I. COMPARISON OF BOTH MODELS BASED ON BLEU SCORE

Papers	Model	*Encoder*	*ROUGE Score*
Ours	Attention LSTM	DenseNet-121	0.272
[2]	LSTM	ResNet-101	0.371
[5]	Attention LSTM	VGG16	0.440
[7]	LSTM	ResNet-152	0.380
[9]	CDGPT2	DenseNet-121	0.281

The score which is taken into consideration here is the recall of the ROUGE-L Score [16]. Recall is the ratio of total number of n-grams in the reference and the number of overlapping n-grams found in both the model output and the reference.

The ROUGE Score obtained is comparable with the other existing work and the score can be improved with better resources available like a larger dataset which would help train the model better thus improving the accuracy and provision of more X-ray images with diseases as most of the images here showed no diseases.

Using much more advanced techniques can also help to obtain a better score. These techniques could be either using Transformers or using Bidirectional Encoder Representations from Transformers (BERT).

V. RESULTS

The performance of each model has been depicted in the tables below. Cumulative N-Gram Scores (BLEU Score) [17] and ROUGE Scores have been used here.

TABLE II. COMPARISON OF BOTH MODELS BASED ON BLEU SCORE

Model	*BLEU - 1*	*BLEU - 2*	*BLEU - 3*	*BLEU - 4*
Encoder-Decoder	0.165	0.113	0.115	0.127
Attention	0.299	0.225	0.226	0.259

TABLE III. COMPARISON OF BOTH MODELS BASED ON ROUGE SCORE

Model	*ROUGE- 1*			*ROUGE - L*		
	r	p	f	r	p	f
Encoder-Decoder	0.105	0.400	0.166	0.105	0.400	0.166
Attention	0.272	0.692	0.391	0.272	0.692	0.391

r - Recall, p - Precision, f - F1 Score

According to the BLEU score and the precision gained from the ROUGE Score, the Attention Model provides superior accuracy. When compared to the Encoder-Decoder Model, the Attention Model's recall and F1-Score are also significantly greater.

Individual N-Gram scores are generated and weighted using the weighted geometric mean for all orders from 1 to N to provide a cumulative N-Gram score. BLEU-1 is a 1-gram score with the weight distribution as (1, 0, 0 , 0) i.e. 100\% of the weight being assigned to the 1-gram score. Similarly, BLEU-2, BLEU-3 and BLEU-4 are 2-gram, 3-gram and 4-gram scores respectively with weight distribution as (0.5, 0.5, 0, 0), (0.33, 0.33, 0.33, 0) and (0.25, 0.25, 0.25, 0.25).

The number of matching 1-grams between our model-generated text and the reference is measured by ROUGE-1, while the longest common subsequence (LCS) between our model output and the reference is measured by ROUGE-L.The precision in ROUGE Score is the ratio of the number of n-grams detected in the model and reference to the total number of n-grams in the model.The F1 Score is calculated with the help of recall and precision.

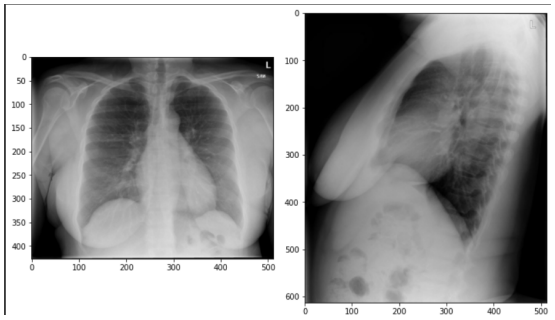

Fig. 2. Frontal and Lateral Chest X-ray

TABLE IV. ENCODER DECODER MODEL RESULTS

Actual	<start> the lungs are clear bilaterally specifically no evidence of focal consolidation pneumothorax or pleural effusion cardio mediastinal silhouette is unremarkable visualized osseous structures of the thorax are without acute abnormality <end>
Predicted	<start> no lungs focal pleural lobe size is in size is in size
BLEU Score	0.213
ROUGE - 1 Score	r - 0.206, p - 0.666, f - 0.315
ROUGE - L Score	r - 0.172, p - 0.555, f - 0.263

TABLE V. ATTENTION MODEL RESULTS

Actual	<start> the lungs are clear bilaterally specifically no evidence of focal consolidation pneumothorax or pleural effusion cardio mediastinal silhouette is unremarkable visualized osseous structures of the thorax are without acute abnormality <end>
Predicted	<start> the heart is normal in size and contour there is no focal airspace opacity no pleural effusion or pneumothorax is seen <end>
BLEU Score	0.301
ROUGE - 1 Score	r - 0.344, p - 0.500, f - 0.408
ROUGE - L Score	r - 0.276, p - 0.400, f - 0.327

VI. Limitations

The size of the dataset in this study is very limited hence model predictions could be improved by adding more data. Additionally, most of the data available in the dataset belonged to the “no disease” category. This could be tackled by obtaining more chest X-ray images with diseases.

VII. Conclusion

It can be clearly stated that the score obtained by the Attention model is higher compared to the Encoder-Decoder model thus proving the efficiency of the Additive Attention Mechanism. As per the resources available, the results may vary and upon further investigation it can be said that there are various other techniques available for report generation which give different accuracies. The application of this project is to act as an aid in becoming a decision support system for the radiologists. The project's long-term goal is to utilize much more modern approaches to produce better results, as well as to create an android or desktop application that can be used as a presentation tool for the generated reports by submitting X-ray images from the radiologist.

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